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JOJEN

Balancing chemical and electrochemical equations

Introduction

Chemical equations is one of the most fundamental things in chemistry, but they are also important in other technical fields and within energy technology - in particular when fuel combustion is considered. Before a chemical reaction can be applied quantitatively to calculate energy released or consumed, it must be balanced correctly. The underlying principle is simply that the same number of each element must be at each side of the arrow between left and right side.

This arrow may take different forms, but this does not have any influence on the balancing of the equation. The double arrow, ⇄, illustrates that the reaction is an equilibrium and that the process can proceed in both directions (from left to right and from right to left). This is always the case in chemistry although often one direction is dominating (perhaps completely dominating). It is in many cases assumed that the process proceeds from left to right and this can be expressed more clearly by a one-way arrow, →.

In the following examples of how to balance various kinds of reactions are given.

Reaction with uncharged molecules

Example 1: Ammonia synthesis from nitrogen and hydrogen

Unbalanced: $N_2 + H_2 \rightleftarrows NH_3$

Start by looking at atoms other than H and O. N_2 means two N and that requires two NH_3 on the right side. $2NH_3$ involves $2 \cdot 3 = 6$ H or 3 H_2.

Balanced: $N_2 + 3H_2 \rightleftarrows 2NH_3$

Control: two N and six H on each side.

Knowledge about which reactions take place when certain reactants react is for chemists, and non-chemists may not know that. However, when it comes to combustion of a fuel, deep chemical knowledge or experience is not necessary. Most fuels are molecules made from carbon and hydrogen and perhaps oxygen. When the fuel is combusted, all carbon atoms form CO_2 and all hydrogen atoms form H_2O.

Example 2: Combustion of methane (CH_4)

Unbalanced: $CH_4 + O_2 \rightleftarrows CO_2 + H_2O$

Look at the carbon first. One CH_4 forms one CO_2 and four H form two H_2O.
$CO_2 + 2H_2O$ on the right side contain four O jointly, which gives us two O_2 to place on the left side.

Balanced: $CH_4 + 2O_2 \rightleftarrows CO_2 + 2H_2O$

Control: one C, four O and four H on each side.

Example 3: Let us try combustion of stearic acid in candle lights. Stearic acid is $C_{18}H_{36}O_2$.

Unbalanced: $C_{18}H_{36}O_2 + O_2 \rightleftarrows CO_2 + H_2O$

Starting with carbon: 18 C give 18 CO_2.
36 H gives 18 H_2O

Partly balanced: $C_{18}H_{36}O_2 + XO_2 \rightleftarrows 18CO_2 + 18H_2O$

We now need $18 \cdot 2 + 18 = 54$ O, but we already have two O in the stearic acid. Consequently, we need to have 52 more O on the left side or 26 O_2

Balanced: $C_{18}H_{36}O_2 + 26O_2 \rightleftarrows 18CO_2 + 18H_2O$

Control: 18 C, 54 O and 36 H on each side.

Reactions involving ions

If ions are involved, the charge must additionally be balanced on each side.

Example 4: Precipitation of sodium sulphate (could be from an aqueous solution where the water evaporates)

Unbalanced: $SO_4^{2-} + Na^+ \rightleftarrows Na_2SO_4$

This is simple. The product on the right side contains two Na and one sulphate, SO_2. Balancing only requires two Na^+.

Partly balanced $SO_4^{2-} + 2Na^+ \rightleftarrows Na_2SO_4$

Atoms fit, but what about the charge? The charge fits too: two minus and two plus equal zero, and therefore it is the final balanced equation.

Balanced $SO_4^{2-} + 2Na^+ \rightleftarrows Na_2SO_4$

Control: just made above.

In this case we ended with zero charge. That is not always the case.

Example 5: Aluminium ions forming aluminium hydroxide ions in basic solution.

Unbalanced: $Al^{3+} + OH^- \rightleftarrows Al(OH)_4^-$

One Al^{3+} form one $Al(OH)_4^-$. Then we just need to balance with XOH^-. This can be done either by looking at the number of O or at the charge. If we look at the charge this time, we must balance 3+ with 1- , and this is done by 4- on the left side (i.e., $4OH^-$)

Balanced $Al^{3+} + \underline{4}OH^- \rightleftarrows Al(OH)_4^-$

Electrochemical reactions

Electrochemical reactions or electrode reactions involve ions and in addition electrons. If the electrons are on the left side of the arrow, they are supplied by the electrode. If they are on the right side, they are consumed by the electrode (assuming the reaction proceeds from left to right). All the criteria above for correct balancing apply, and now we will also add electrons. The number of electrons must be the same on each side, but that is not easy to check since most electrons are implicitly embedded in the molecules. However, this is not a problem, because as long as the charge is balanced, the total number of electrons will be too. The number is given by the charge of the nuclei which are fixed by the atom names. So, make sure the charge is balanced by electrons, and all is fine.

Example 6: Equilibrium between metallic copper and copper ions.

Unbalanced: $Cu \rightleftarrows Cu^{2+}$

The atoms are already balanced, so only the charge remains to be balanced. This is done by two electrons on the right side.

Balanced: $Cu \rightleftarrows Cu^{2+} + 2e^-$

Example 7: Electrochemical oxidation of methanol, CH_3OH. In this case oxidation is done by exchange of electrons with an electrode instead of oxygen.

Unbalanced: $CH_3OH + H_2O \rightleftarrows CO_2 + H^+$

One CH_3OH gives one CO_2. All hydrogen atoms become H^+ (protons i.e., hydrogen ions), but we do not know how many before the number of H_2O is settled. This is done by the oxygen balance: One CO_2 demands two O. One is already in CH_3OH, so we need only one H_2O. The positive charge of the protons must be balanced by the negative charge of the electrons that we add on the same side in this case.

Partly balanced: $CH_3OH + H_2O \rightleftarrows CO_2 + XH^+ + Xe^-$

There are six hydrogen on the left side, so we have $6H^+$ on the right and thus also $6e^-$.

Balanced: $CH_3OH + H_2O \rightleftarrows CO_2 + 6H^+ + 6e^-$

The electrons liberated are consumed by the electrode, and the protons proceeds into the electrolyte solution. Such a reaction cannot proceed alone since it would lead to accumulation of negative charge in the electrode and of positive charge in the solution. The process will almost immediately stop for that reason[1] and to proceed further, it must be combined with another electrode process consuming the electrons and the protons. Such an electrode process could be reaction with oxygen and the formation of water as in Example 8.

Example 8:

Unbalanced: $H^+ + O_2 \rightleftarrows H_2O$

One O_2 will form two H_2O and then we need four H^+.

Partly balanced: $4H^+ + O_2 \rightleftarrows 2H_2O$

The charge is then balance by four electrons.

Balanced: $4H^+ + O_2 + 4e^- \rightleftarrows 2H_2O$

Now the equation is correctly balanced, but if we like it to fit the reaction in Example 7, we need to match the number of electrons and protons in the two reactions. One way is to multiply this one by 1.5:

Balanced: $6H^+ + 1.5O_2 + 6e^- \rightleftarrows 3H_2O$

Or we can divide the equation of Example 7 by 1.5:

Balanced: $\frac{2}{3}CH_3OH + \frac{2}{3}H_2O \rightleftarrows \frac{2}{3}CO_2 + 4H^+ + 4e^-$

[1] At the same time, the electrode has reached its electrochemical equilibrium potential as a balance between the ever so small charge imbalance and the chemical driving force for the reaction. However, this is not important in this context.

All the balanced equation are correct. It is a matter of taste which way of writing them to prefer. The combination of the reaction in Examples 7 and 8 is the reaction scheme of a methanol powered fuel cell.

Methanol sub-process: $CH_3OH + H_2O \rightleftarrows CO_2 + 6H^+ + 6e^-$

Oxygen sub-process: $6H^+ + 1.5O_2 + 6e^- \rightleftarrows 3H_2O$

Total: $CH_3OH + 1.5O_2 \rightleftarrows CO_2 + 2H_2O$

In "Total", the two processes above are merged, and one H_2O is removed on each side. We never write the same species on both sides.

Final note

It is a general principle when writing chemical reactions to have only species that react are on the left side and only species formed on the right side. If we work with. When addressing ionic species in solution

Silver nitrate ($AgNO_3$) is highly soluble, and silver chloride (AgCl) is not. The reaction between $AgNO_3$ and NaCl is

Correct: $AgNO_3 + NaCl \rightleftarrows AgCl + NaNO_3$

The species $AgNO_3$ + NaCl react and are consumed and the species AgCl and $AgNO_3$ are formed. If we write the same reaction with the ions in solution we get

Incorrect: $Ag^+ + NO_3^- + Na^+ + Cl^- \rightleftarrows AgCl + Na^+ + NO_3^-$

This is balanced correctly by atoms and charge, but the species Na^+ and NO_3^- are not part of the reaction, so they shall be omitted [2]

Correct: $Ag^+ + Cl^- \rightleftarrows AgCl$

In other words, after the salts are dissolved, only silver and chloride are relevant to the reaction.

[2] In some cases, however, when a fuel is combusted in air and not pure oxygen. It may be practical to include all species present. If temperature changes or pressure changes are to be calculated, all species are relevant and it makes more sense to include all species. The case for methane combustion is then $CH_4 + 2O_2 + 8N_2 \rightleftarrows CO_2 + 2H_2O + 8N_2$. This is not a correct balancing, but a practical trick.